alongside manned aircraft. In civilian sectors, autonomy enables drones to perform precision agriculture, infrastructure inspection, and real-time traffic monitoring, while consumer-level drones use automated flight tracking to capture complex imagery without skilled piloting.

However, the rise of autonomous drones also brings forward important ethical, legal, and regulatory considerations related to safety, accountability, privacy, and airspace management. Ensuring responsible and secure deploy ment requires well-defined protocols, intelligent traffic control systems, and transparent human oversight frameworks. As AI continues to advance, drones will increasingly function as self-adapting aerial agents within interconnected digital ecosystems, marking a decisive step toward the development of intelligent, collaborative, and fully integrated aerial infrastructures of the future.

9.2.1 Full autonomy: Drones making real-time decisions without human intervention

Full autonomy in drones refers to the ability of UAVs to conduct entire missions independently, without the need for direct or continuous human control. This capability is enabled through the integration of advanced onboard sensors, real-time data processing, and AI-driven decision-making algorithms, which together allow the drone to perceive its surroundings, analyze situational variables, and adapt its actions accordingly. Fully autonomous drones can take off, navigate, avoid obstacles, carry out assigned tasks, and land safely even in unfamiliar or dynamic environments.

Such autonomy is especially valuable in high-risk or inaccessible areas, including disaster zones, remote terrain, deep forests, conflict regions, and offshore industrial sites, where human piloting may be dangerous or impractical. The drone continuously interprets input data from visual,